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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

REVISED SUBMISSION COPY - leader review/signature pending.

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Phase 1: Preliminary Project Proposal

1. Team & Project Overview

Field	Response
Team Name	奶龙 (nailong)
Selected Domain	Domain 2
SDG Track	SDG 8 - Decent Work and Economic Growth; targets 8.2 / 8.3
Project / Solution Name	Nexus
Leader	LING YUXUAN
Email & Phone (Leader)	1730211684@qq.com 1123232813
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context (maximum 250 words)

Small print and custom-gift businesses receive customer enquiries through messages while stock, staff availability and task ownership sit in separate records. Customer service staff or the owner repeatedly re-enter requirements, check who is available and ask colleagues for progress. Missing information and unclear handoffs can lead to premature delivery promises or requests with no accountable owner.

Nexus focuses on this recurring request-to-task burden for small Malaysian teams, initially targeting print and custom-gift businesses. This is a design hypothesis derived from the challenge brief and prototype scenario, not a completed customer-research finding. It matters economically because avoidable checking and follow-up consume limited staff time. The project prioritises productivity and accessible workflows rather than claiming measured financial or environmental savings. The initial workflow is designed for mixed digital confidence, a mobile interface, RM amounts and Malaysia time.

2.2 The Proposed AI Solution (maximum 300 words)

Nexus turns a customer message into a checked, human-approved task. Customer service confirms requirements; fixed rules inspect synthetic stock, staff skills and approved leave, identify missing information or conflicts, and suggest an assignee. An authorised demo manager reviews the proposal. Approval rechecks current data and prevents repeated execution before creating a linked task and activity record.

A deployed server-side OpenAI Responses API adapter can request structured extraction and an editable reply draft from GPT-4.1 mini. It processes only the message explicitly selected by staff and never approves work or sends a reply automatically. Live verification reached OpenAI but returned credit_balance_exhausted, so successful AI generation has not yet been verified. The manual and rules-based workflow remains usable.

Local Excel (.xlsx) and CSV employee-resource tables can be analysed without an AI call or Microsoft 365 account. Business data and demo roles remain browser-local; WhatsApp and online Microsoft workbook synchronisation are not connected. People, Work and Connect supply workflow context. Phase 2 will prioritise a narrow, tested handoff workflow and authorised integrations instead of claiming a complete enterprise platform.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack and Architecture

Nexus is deployed on OpenAI Sites as a Cloudflare Worker serving the frontend and backend endpoints. Browser-side rules check synthetic stock, staff availability and approved leave. Demo roles review proposals before linked tasks and activity records are created. Business records remain in browser localStorage.

For AI assistance, customer service explicitly submits one selected message to a same-origin backend endpoint. The backend requires ChatGPT sign-in, enforces daily usage limits in Cloudflare D1 and calls OpenAI Responses API using a server-side secret. Structured outputs and source evidence are validated before editable suggestions are returned. AI does not send replies, approve requests or create tasks automatically. Live verification reached OpenAI but was blocked by insufficient API credits.

Excel/CSV employee tables are read locally in the browser. WhatsApp and Microsoft 365 cloud synchronisation are not connected. Replit sponsor-provided premium access was used for development and testing; it is separate from the current Sites deployment.

Layer	Technology / current role
Frontend	HTML5, CSS, native JavaScript; fflate 0.8.3 for local .xlsx decompression.
Backend / hosting	Cloudflare Worker via OpenAI Sites; server-only OpenAI secret.
Storage	localStorage for demo business records; D1 for AI counters only.
Testing	Node.js 24; 61 automated tests plus browser checks.
Sponsor resource	Replit SDC-provided premium access for development/testing. No sponsor-provided GenAI credits are confirmed.

3.2 Design-Thinking & Deployment Strategy (maximum 200 words)

We use understanding, prototyping, testing and iteration to focus on repeated request checking and handoff. The prototype makes each role's next action explicit and keeps consequential decisions with people. Sixty-one automated tests cover business rules, access-scope demonstrations, approval checks, AI input/output boundaries and Excel import validation. Browser checks verified representative workflows, real file reading and clear API failure feedback.

Before Phase 2, we plan to observe at least five target users completing matched tasks, record completion time, missing information and requests for help, then simplify the interface. We will replace the estimated manual baseline with observed timings. Real enterprise trials require server-side company membership and tenant authorisation, transactional business storage and recoverable external integrations. We will connect only an authorised business channel and validate one end-to-end scenario before widening scope. These are readiness actions, not completed production capabilities.

4. Preliminary Environmental & SDG Impact Framework

Metric Category	Response
Target SDG sub-targets	SDG 8.2: productivity through technological upgrading and innovation. SDG 8.3: support for productive activities and MSME growth. These are alignment goals, not claims of measured national impact.
Current problem baseline	Initial team estimate: 10 minutes of active staff handling per request - 3 minutes to structure requirements, 4 to check stock and staff, and 3 to assign and record the work. This is not an industry statistic or a measured pre-prototype baseline. It will be replaced with an observed manual baseline before impact claims are made.
Core measurable KPIs	Median active staff minutes per request; percentage of requests missing required information; and percentage of critical actions with authorised human approval. Record failed tasks and sample size, not only successful completions.
Phase 2 impact target	Phase 2 target: reduce median active staff handling time by 30% versus the measured manual baseline (illustratively 10 to 7 minutes), while maintaining 100% human approval coverage for critical actions. Plan: at least five target users and 30 matched manual/Nexus task pairs. These are proposed evaluation targets, not achieved results.

Source for SDG definitions: United Nations, Goal 8, targets 8.2 and 8.3: https://sdgs.un.org/goals/goal8 . The 10-minute baseline is a transparent team workflow estimate; no external source is claimed for it. No unsupported 49% / 59% figures or achieved cost/waste reductions are used.

5. Eligibility & Readiness Checklist

[x] All team members completed the mandatory GenAI, Database and Cloud Deployment workshops (team-confirmed).
[ ] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs (leader to confirm the validation basis).
[x] A functional deployed prototype exists, with API-credit and production-readiness limits disclosed.
[ ] Leader to confirm all final fields, word limits and official submission requirements after reviewing this revised copy.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

No, as declared by the team in the supplied proposal. The team must verify this remains accurate for all members. No additional prior-owned work has been asserted in this revision.

6.2 Third-Party Code, Datasets, or Models

Yes. The following third-party resources are used or called by the implementation; development assistance is separately declared in the technical declaration.

Name	Type	Licence / terms
fflate 0.8.3	Code library	MIT; full attribution retained.
Node.js v24.19.0	Development runtime	MIT and bundled third-party notices.
GPT-4.1 mini	Pre-trained API model	Proprietary; OpenAI API service terms. Live generation is credit-blocked.

6.3 Employment-Related IP Conflicts

No, as declared by the team in the supplied proposal. Team members remain responsible for checking applicable employment, internship and consulting agreements.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team knowledge, and that the project will be original work created during the SDC 2026 hackathon period.

Leader review and signature are still required. No signature is applied by this revision.

Field	Response
Leader	LING YUXUAN
Signature	Leader to sign after final review.
Prepared on	24 October 2026

Appendix A - Current End-to-End System Blocks

Path	Connected components
Business workflow	Customer/support UI -> confirmed fields -> local stock/staff/leave rules -> human approval -> tasks and local activity records.
AI assistance	Explicitly selected message -> Sites Worker -> ChatGPT identity and D1 quota check -> OpenAI Responses API -> validated, editable draft -> human review.
File import	Local .xlsx / .csv -> browser parser (fflate for .xlsx) -> field and row validation -> employee-resource summary. File bytes are not uploaded.
Cloud runtime	Sites deployment -> Cloudflare Worker serving UI and API; D1 persists usage counts only; OPENAI_API_KEY remains in server secrets.
Current limits	API balance blocked successful generation. Business roles and records remain browser-local. WhatsApp and Microsoft 365 synchronisation are unconnected.

Team Members

Name	Confirmed capacity
LING YUXUAN	Team leader
LI JUNZHANG	Team member
XIE WEIYE	Team member
HU ZHAOMING	Team member

The two official PDF templates do not ask for individual duty assignments. No unconfirmed duties are invented here. The repository may separately record team roles once confirmed.

References and Submission Status

United Nations SDG 8: https://sdgs.un.org/goals/goal8
Official repository template: https://github.com/michaelzhngg/SDC_Hackathon_Submission_Template
Application branch: https://github.com/junzhanglee/SDC2026-domain2-nailong/tree/feat/nexus-mvp
Live prototype: https://nexus-nailong-2026.d7ystszcjf.chatgpt.site/

Outstanding submission actions: leader review/signature; actual Replit proof-of-use evidence; slide-deck and demo-video links; organiser confirmation of RunPod proof applicability. This review copy does not fabricate these items.